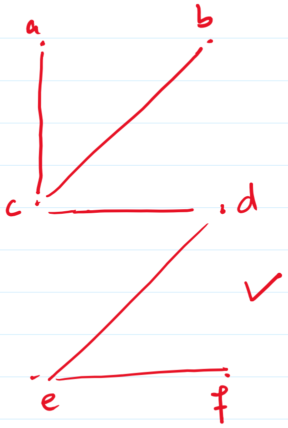
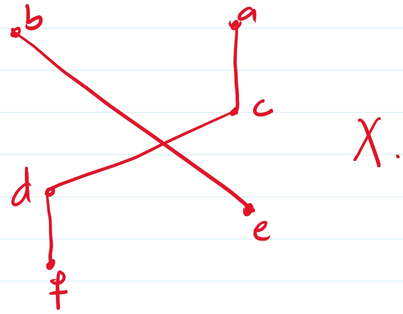
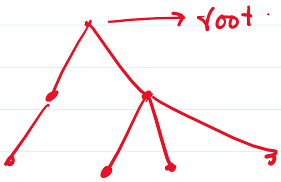
Lecture 22. TREES.
- Connected
- Undirected.
- No simple Circuit.





Rooted Tree :- One Vertex designated as a root. and every edge is directed away from it.

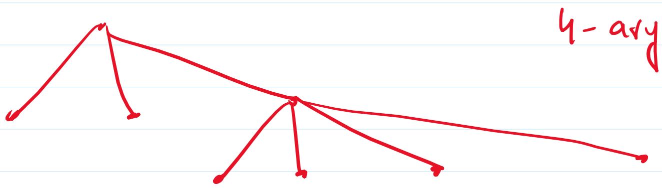


1) Parent
2) Child.
3) Sibling.

4) Ancestors.
5) Descendents.
6) Internal Vertex.

7) Leaf.

m-ary Tree.

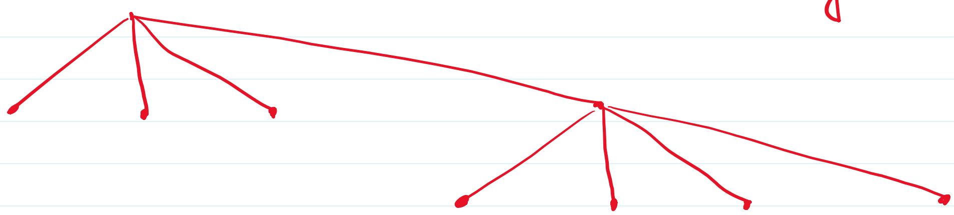


4-ary Tree.

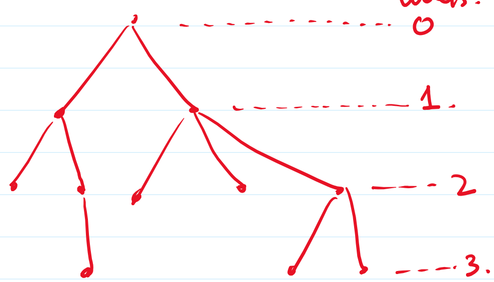
Full m-ary Tree.

Full m-ary Tree.

Full 4-ary tree.



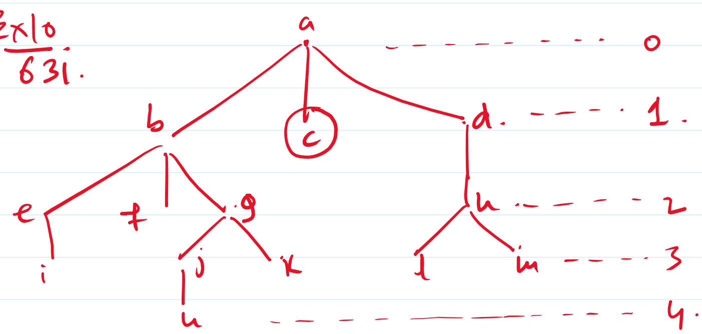
Levels.



h = height = highest level of any Vertex = 3.

Balanced Tree.

Ex 10
631.



h = 4.
h − 1 = 3.

X.

theorem :- A tree with n Vertices has n−1 edges.

theorem :- A full-m ary tree with i Internal Vertices has n = mi+1 Vertices.

Applications of Trees.

Binary Search Tree.

Ex 1 :-
P636

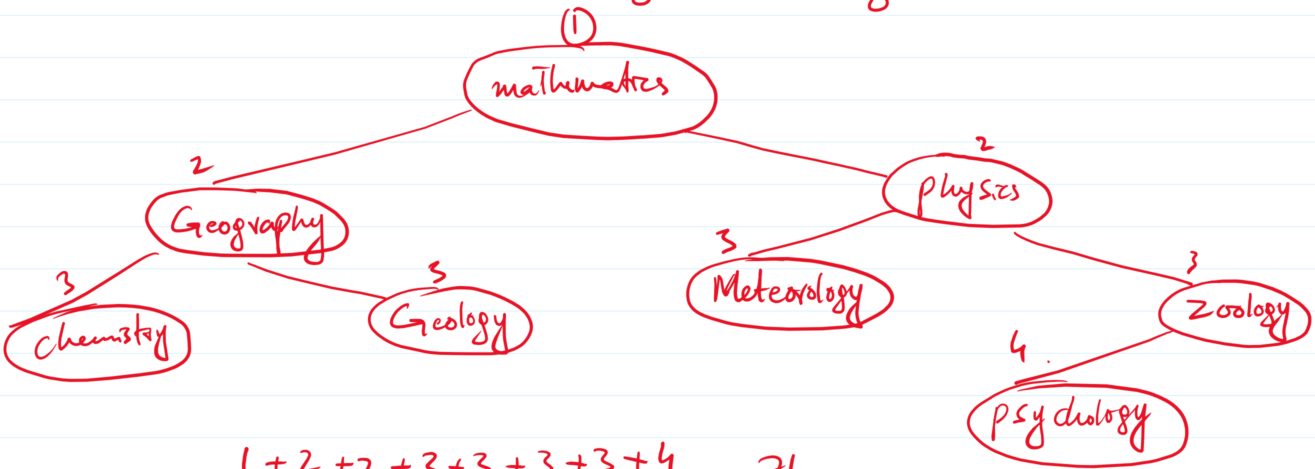
[Mathematics, Physics, Geography, Zoology, Meteorology, Geology, Psychology, Chemistry].

Ex1 :-
P636

[Mathematics, Physics, Geography, Zoology, Meteorology, Geology, Psychology, Chemistry].

(The digits 5, 6, 7 and 8 are written above Meteorology, Geology, Psychology and Chemistry.)

$$\frac{1+2+3+4+5+6+7+8}{8} = \frac{36}{8} = 4.5$$



$$\frac{1+2+2+3+3+3+3+4}{8} = \frac{21}{8} = 2.625.$$

Huffman Coding.

a	000
b	001
c	010
d	011
e	100
f	101.

$\boxed{\text{aaabcd cbaaa eefab.}}$

	a	b	c	d	e	f
freq.	7	3	2	1	2	1.
prob	7/16	3/16	2/16	1/16	2/16	1/16.

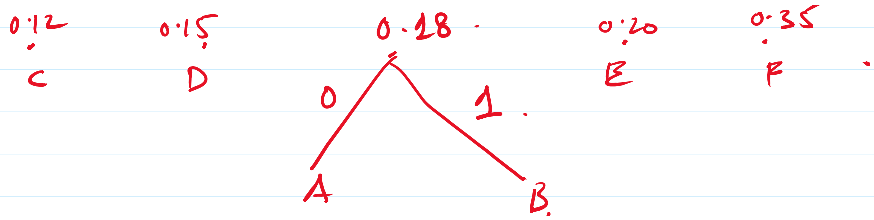
Avg Bits per letter $= 0.437\times3 + 0.187\times3 + 0.125\times3 + 0.0625\times3 + 0.125\times3 + 0.0625\times3$

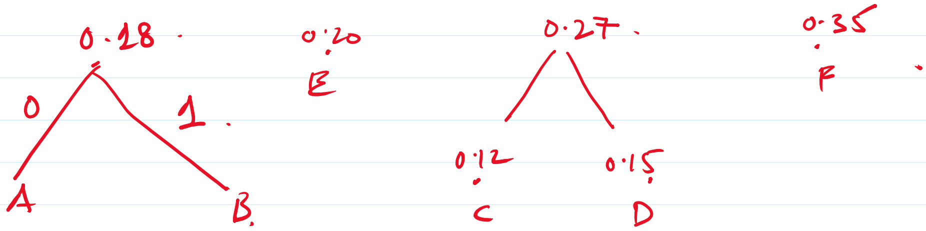
Avg Bits/letter $= 3$. bit/letter.

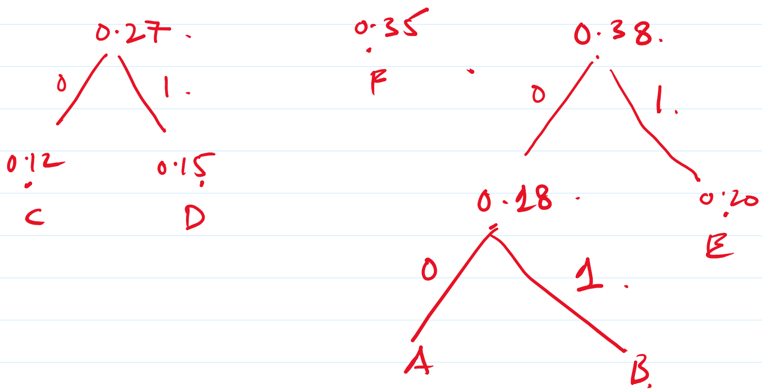
Ex5 :-
640

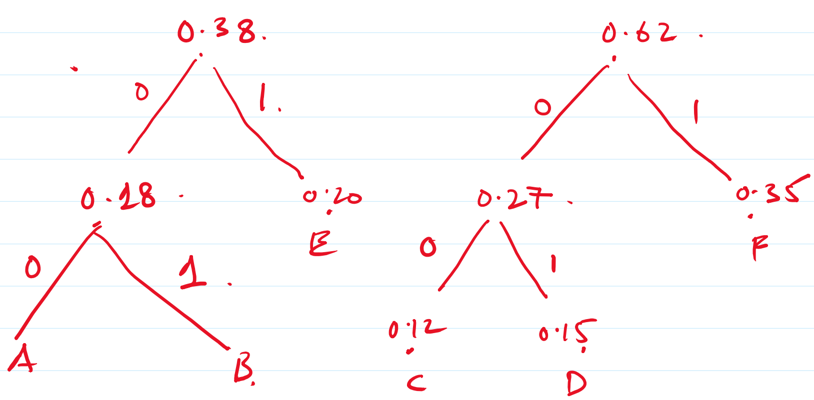
0.08	0.10	0.12	0.15	0.20	0.35
A	B	C	D	E	F.

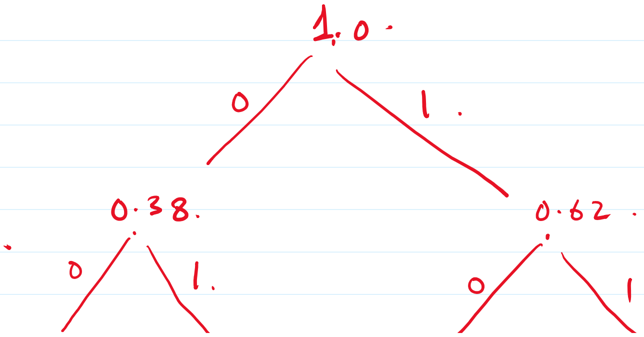
0.12	0.15	0.18	0.20	0.35
C	D	0 1	E	F.



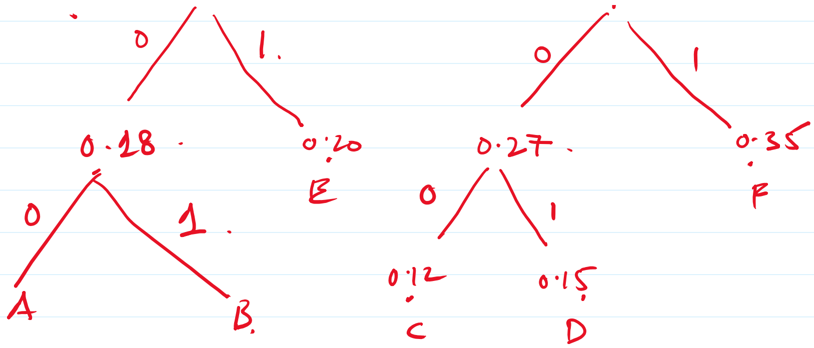








A = 000
B = 100
C = 001
D = 101
E = 10
F = 11



D = 101.
E = 10.
F = 11.

$$\text{Avg bits/letter} = 0.08\times 3 + 0.1\times 3 + 0.12\times 3 + 0.15\times 3 + 0.2\times 2 + 0.35\times 0.2$$
$$= 2.45 \text{ bit/letter}.$$